



Vehicle detection and classification by measuring and processing magnetic signal

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ABSTRACT

This paper presents novel vehicle detection and classification method by measuring and processing magnetic signal based on single micro-electro-mechanical system (MEMS) magnetic sensor. When a vehicle moves over the ground, it generates a succession of impacts on the earth's magnetic field, which can be detected by single magnetic sensor. The magnetic signal measured by the magnetic sensor is related to the moving direction and the type of the vehicle. Generally, the recognition rate using single sensor detector is not high. In order to improve the recognition rate, a novel feature extraction algorithm and a novel vehicle classification and recognition algorithm are presented. The concavity and convexity areas, and the angles of concave and convex parts of the waveform are extracted. An improved support vector machine (ISVM) classifier is developed to perform vehicle classification and recognition. The effectiveness of the proposed approach is verified by outdoor experiments.

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1. Introduction

In recent years, Intelligent Transportation System (ITS) programs have been initiated in many countries [1–3]. As a part of ITS, moving vehicle detection and classification system plays an important role since it gathers all parts of the data used in ITS. Video image, acoustic, seismic, and infrared sensors are used in vehicle detection and classification systems. However, the detection and classification systems based on video image, acoustic, seismic and infrared sensors are expensive and large in size due to the usage of extra hardware. Furthermore, their performance is often affected by Doppler effects and noise introduced by moving vehicles and atmospheric and terrain variations. In contrast, magnetic sensors are less sensitive to noise and Doppler effects. Moreover, magnetic sensors do not require the 'line of sight' assumption and are relatively immune from weather conditions. They can detect objects through obstacles such as foliage. Among various

magnetic sensors, the micro-electro-mechanical system (MEMS) magnetic sensors have advantages in terms of cost, size, weight and power consumption [4–7]. Hence, applying MEMS magnetic sensors to vehicle detection, classification and recognition is a promising research topic.

Currently, some methods on vehicle detection and classification using magnetic sensors have been reported in literature. These methods are either based on the fusion of magnetic and other kinds of sensing techniques or based on the fusion of magnetic arrays. Ding et al. proposed a vehicle detection algorithm called the Adaptive Threshold algorithm based on acoustic sensor and magnetic sensor [8]. In [9], Casalegno proposed a vehicle classification method based on magnetometer arrays. Their recognition rates are between 80% and 90%. Since multiple sensors and fusion techniques are used, these methods are usually complicated.

In this paper, a new approach is proposed to detect and classify vehicles by measuring and processing magnetic signal based on single MEMS sensor. It is superior to the existing methods in terms of reliability, installation and maintenance. The feasibility of the approach is based on the magnetic signal which is closely related to the

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characteristics of the vehicle. By exploiting the features of the magnetic signal, our approach can recognize the moving direction and the type of vehicle with high recognition rate. It is also computationally efficient as it uses an improved one-against-all classification algorithm based on SVM. Outdoor experiments have been conducted to test the performance of the new approach.

2. Foundation of vehicle detection and recognition using single magnetic sensor

The sensor considered in this paper is the 1-axis MEMS magnetic sensor from Honeywell [10]. It is made of Permalloy (NiFe) thin film deposited onto a silicon substrate and patterned to form a Wheatstone resistor bridge (see Fig. 1) which can generate a differential output voltage. In the bridge, the opposite elements are identical. If the magnetic field in the sensitive axis increases, the V_b to Out+ and Out– to GND elements will slightly decrease in resistance while the other two elements will increase in resistance. This results in an increase of voltage at Out+ and a decrease of voltage at Out–, or equivalently an increase of voltage from Out+ to Out–. Similarly, if the magnetic field in the sensitive axis decreases, the voltage across Out+ and Out– will decrease. Hence, by measuring the voltage across Out+ and Out–, one can detect the change of the magnetic field in which the sensor is placed.

The magnetic field disturbance produced by a moving vehicle can be measured by a single axis magnetic sensor. The distortion in the earth's magnetic field concentrates mainly at locations near the engine and wheels. However, it can also vary depending on the internal distribution of the ferrous materials used in other parts of the vehicle, such as rooftop and trunk locations. The net result is a characteristic distortion to the earth's magnetic field that is unique to the shape of the vehicle. To verify this view point, we used a

MEMS magnetic sensor to detect typical vehicle targets. The measurement system is shown in Fig. 2 which includes a power unit, a MEMS magnetic sensor and conditioning circuit, a high-speed A/D–D/A data acquisition card and a computer. The sensor was installed by the side of a road and the vehicles ran beside it (see Fig. 3).

The test was carried out on a straight concrete road that is 2-km long and 12-m wide. The vehicles tested include heavy tracked vehicles, light tracked vehicles and light wheeled vehicles. It was a sunny day with 30–35 °C temperature and 1–2 wind force level. The sensor is installed in the way that its sensitive axis is parallel with the directions of moving vehicles. The direction of sensitive axis of the sensor is called “forward direction” and the reverse direction of sensitive axis is called “reverse direction”. The speeds of target vehicles were not the same but each vehicle passed by the sensor with a uniform speed. Also, the vehicles moved passed the sensor within a distance of three meters to ensure that the vehicles were in the detection range of the sensor. The sensor signal was sampled at the frequency of 2000 Hz.

Fig. 4 shows the signal waveforms generated by a heavy tracked vehicle traveling in different directions. It can be seen from Fig. 4b that when there is no vehicle in the sensor's detection range, the output of the sensing circuit is initially zero. When the vehicle is traveling in forward direction, the output will first decrease below the initial value. Then it increases and reaches the initial value at the time when the vehicle is in front of the sensor. When the vehicle moves further, the flux lines will bend toward the vehicle. As a result, the signal value increases and becomes positive. It will return to the initial value again when the vehicle moves out of the sensor's detection range. Fig. 4a shows the signal waveform when the vehicle travels in the reverse direction. It is easy to see that the waveform in Fig. 4a is approximately a mirror reflection

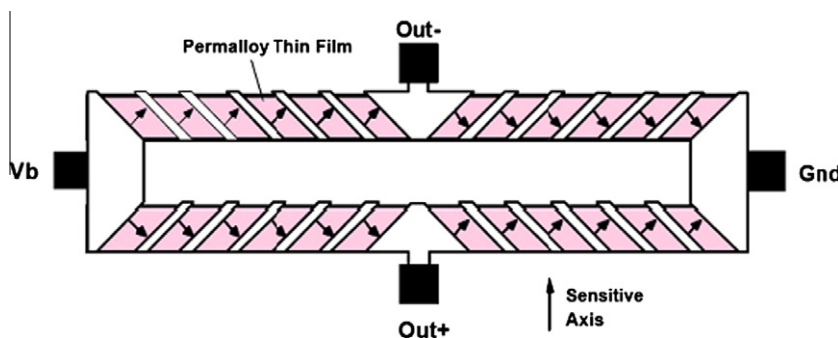


Fig. 1. Magneto-Resistive Wheatstone bridge elements.

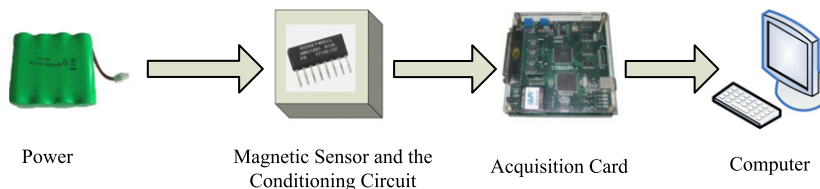


Fig. 2. Structure of the detection system.

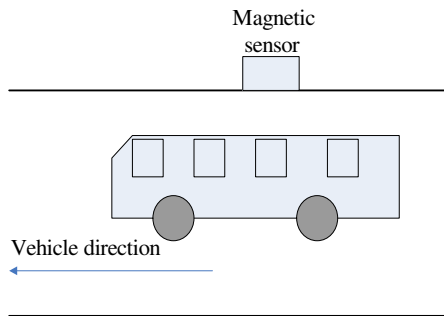


Fig. 3. Sensor installation diagram.

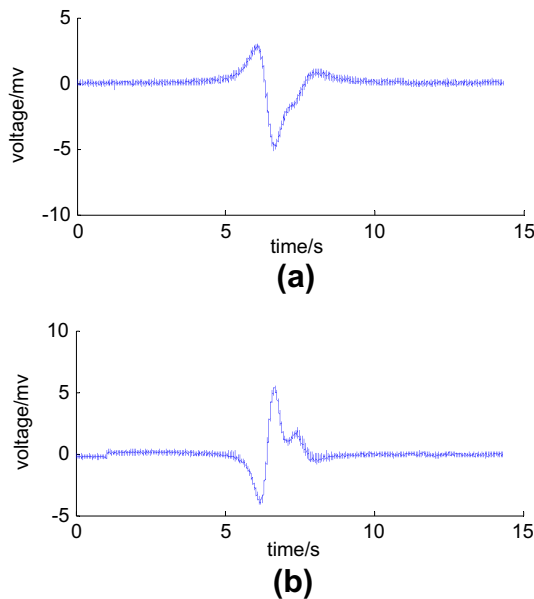


Fig. 4. Magnetic signals of a heavy tracked vehicle versus different moving directions: (a) reverse direction and (b) forward direction.

of that in Fig. 4b. Clearly, the polarity of the waveform can be used to detect the moving direction of the vehicle.

Fig. 5 shows the voltage waveforms of the magnetic signals generated respectively by a heavy tracked vehicle, a light tracked vehicle and a light wheeled vehicle, where the vehicles were traveling in the same direction (reverse direction). Clearly, these waveforms are quite different and thus their features can be exploited to classify and recognize the vehicle types. These observations lay a foundation for target classification and recognition. In the remainder of this paper, we shall focus on how to determine the types of vehicles by exploiting the features of the magnetic signals. An efficient algorithm will be proposed to carry out vehicle classification and recognition.

3. Vehicle recognition based on the features of magnetic signals

The vehicle recognition process is shown in Fig. 6. It consists of four steps: noise elimination, signal segmentation, feature extraction, and vehicle classification and recognition.

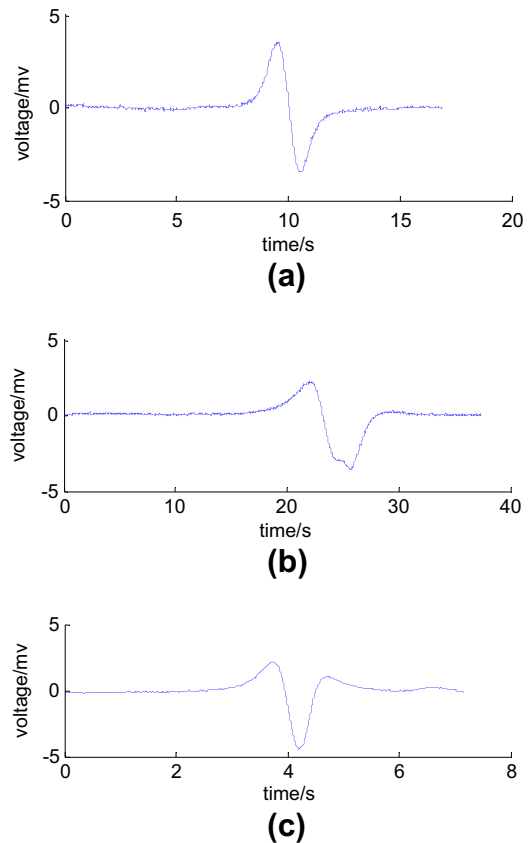


Fig. 5. Magnetic signals generated by different types of vehicles: (a) heavy tracked vehicle; (b) light tracked vehicle and (c) light wheeled vehicle.

3.1. Noise elimination

In some cases, relatively strong additive noise appears and affects the measurements. The noise signals could be caused by environmental factors and/or by the operations of electronic circuits, such as the data transmission from the acquisition card to the host computer. The spectrum analysis shows that the periods of the noise signals encountered in this application are less than several micro seconds. In contrast, as shown in Figs. 4 and 5, the length of a vehicle signal waveform is much greater than the periods of noise signals. As a result, a low-pass filter can be used to effectively remove noise signals. Here we use the digital filtering technique to eliminate noise [11–14]. The effectiveness of this technique can be demonstrated by the example shown in Fig. 7. It is clear that after noise elimination, the useful information of the magnetic signal is well reserved. This technique is applied to filtering magnetic signals generated by various vehicles and satisfactory results are obtained in all cases.

3.2. Signal segmentation

It can be seen from the plots of magnetic signals that only a small portion of the waveform contains useful

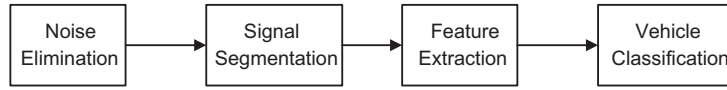


Fig. 6. Flow chart of the vehicle recognition.

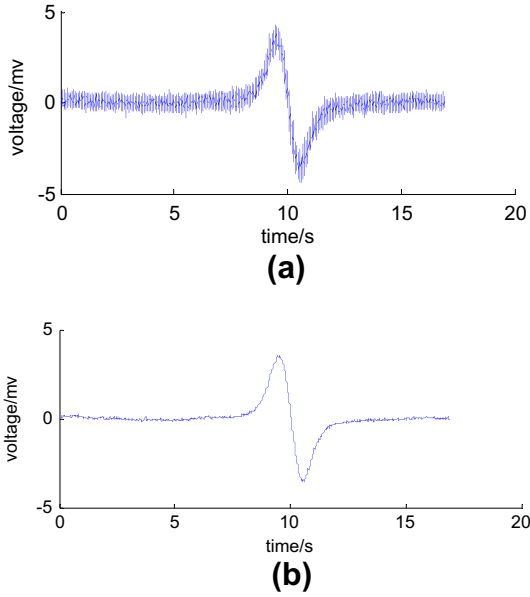


Fig. 7. An example of noise elimination: (a) original signal and (b) signal after noise elimination.

information of the vehicle passing by the sensor. The rest of the waveform is almost static. The static part of the waveform should be removed to save computing time. This can be done by means of signal segmentation. Fig. 8 shows an example of signal segmentation conducted on the magnetic signal from a heavy tracked vehicle.

3.3. Feature extraction

In order to numerically identify the type of a vehicle, some quantitative indices need to be proposed to reflect the features of the magnetic signal. These indices should have high correlations with the shape of the signal waveform and can be easily computed. The feature indices proposed for vehicle classification and recognition are the concavity and convexity areas, and the angles of the concave and convex parts of the waveform. Before introducing the definitions of these feature indices, some important points are labeled on the segmented signal shown in Fig. 8b. The coordinates of the four points corresponding to the beginning, the maximum value, the minimum value and the end of the segmented signal are denoted by (x_1, y_1) , (x_2, y_2) , (x_3, y_3) and (x_4, y_4) , respectively. In practice, the points (x_2, y_2) and (x_3, y_3) can be determined by looking at the maximum and minimum of the signal while points (x_1, y_1) and (x_4, y_4) can be determined by looking at the slope of the signal or its magnitude. Between the points (x_2, y_2) and (x_3, y_3) , there exists one point corresponding

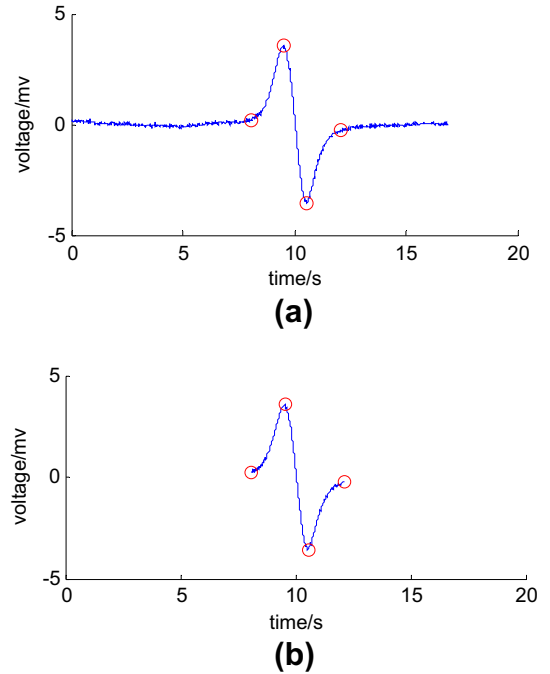


Fig. 8. Signal segmentation for a magnetic signal: (a) signal before segmentation and (b) signal after segmentation.

to the voltage of zero volt which is denoted by (x_5, y_5) . Clearly, $y_5 = 0$.

Based on these notations, the convexity area can be approximately computed through discrete data integral as

$$S_1 \approx \sum_{x_1 \leq i \leq x_5} y_i \quad (1)$$

where $y_i > 0$. Similarly, the concavity area is

$$S_2 \approx \sum_{x_5 \leq i \leq x_4} (-y_i) \quad (2)$$

where $y_i < 0$. Let $K_1 = \frac{(y_2 - y_1)}{(x_2 - x_1)}$, $K_2 = \frac{(y_3 - y_2)}{(x_3 - x_2)}$ and $K_3 = \frac{(y_4 - y_3)}{(x_4 - x_3)}$. Then the angles of the convex and concave parts are defined respectively as

$$\tan(\theta_1) = \left| \frac{(K_2 - K_1)}{(1 + K_2 \cdot K_1)} \right| \quad (3)$$

and

$$\tan(\theta_2) = \left| \frac{(K_3 - K_2)}{(1 + K_3 \cdot K_2)} \right| \quad (4)$$

where, θ_1 represents the convex angle, θ_2 represents the concave angle. Fig. 9 shows the distribution of the concavity and convexity areas of three types of vehicles. The distribution of the angles of concave part and convex part is

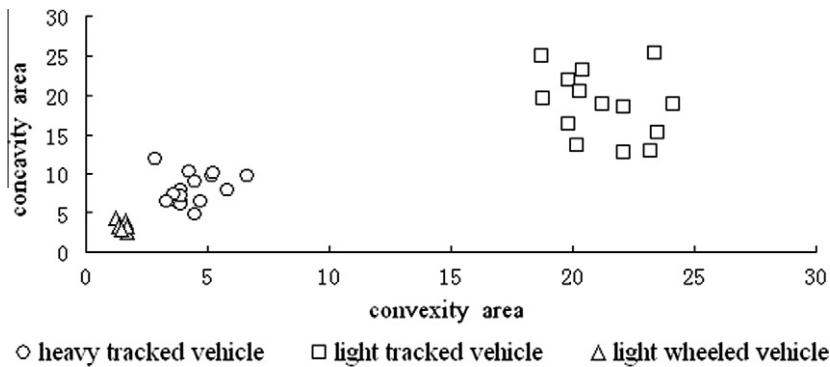


Fig. 9. Distribution of concavity and convexity areas.

shown in Fig. 10. From Fig. 9, it is clear that the convexity areas of light tracked vehicles are much larger than those of heavy tracked vehicles and light wheeled vehicles. By exploiting this property, light tracked vehicles can be easily separated from the others. In addition, as shown in Fig. 10, the angles of convex part of light wheeled vehicles are much smaller than those of light and heavy tracked vehicles. This property can be exploited to separate light wheeled vehicles from the others. Therefore, vehicle classification and recognition can be achieved by jointly exploiting these properties.

3.4. Vehicle classification and recognition

The support vector machine (SVM) is a statistical learning method based on the structure risk minimization principle. Its efficiency has been proved in many pattern recognition applications and can be used to classify and recognize vehicles. To classify more than two types of vehicles, a multi-class SVM must be used. The two widely used multi-class SVMs are the one-against-all method and the one-against-one method [15–17]. In the one-against-all method, a set of binary classifiers are first trained to be able to separate each class from the others. Then each data object is classified to the class for which the largest decision value was determined. To classify M classes, this method needs M classifiers. Differently, in the

one-against-one method, a series of classifiers is applied to each pair of classes, with the most commonly computed class kept for each object. Then a max-win operator is used to determine to which class the object will be finally assigned. This method requires $\frac{M(M-1)}{2}$ classifiers. So the one-against-all method is, in general, superior to the one-against-one method as it requires much less classifiers.

However, the one-against-all method is computationally demanding. The reason is that in the testing process, all M classifiers must be used to determine the class of a data object. Another drawback of the method is that every data object will be classified to one of the M classes. If a data object does not belong to any of the M classes, a classification error will occur. Next, we shall propose an improved one-against-all algorithm that overcomes these drawbacks. The classifier based on the improved one-against-all algorithm is named as improved SVM classifier, or ISVM classifier.

We first consider the construction of the M binary SVM classifiers using N training samples $\{x_1, y_1\}, \dots, \{x_N, y_N\}$. Here, $x_i \in R^k$ is a k -dimensional feature vector corresponding to the i -th training sample, and $y_i \in \{1, 2, \dots, M\}$ is the class label of x_i . When constructing the m -th classifier, we re-write the class label of x_i to be y_i^m , where $y_i^m = 1$ if $y_i = m$ and $y_i^m = -1$ if $y_i \neq m$. Then based on the concept of the two-class SVM method, the decision function for the m -th classifier is

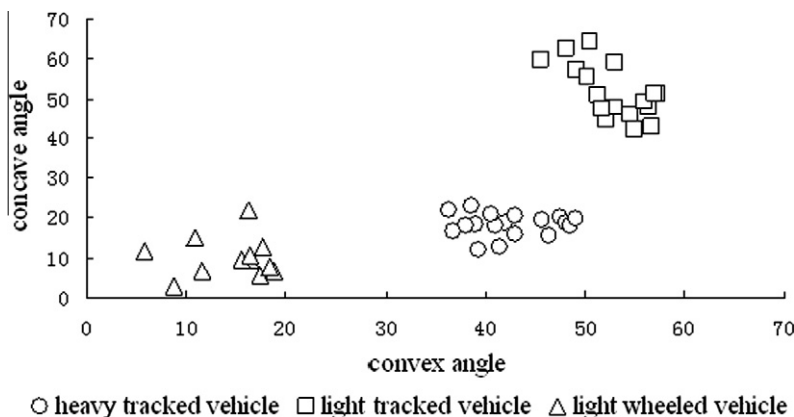


Fig. 10. Distribution of the angles of concave part and convex part.

$$f^m(x) = \text{sign} \left(\sum_i a_i^m y_i^m K(x_i, x) + b^m \right) \quad (5)$$

where, a_i^m is a weighting factor and $a_i^m \geq 0$, b^m is a bias and $K(x_i, x)$ is a known symmetric positive function called kernel. Since $y_i^m (w_i^T x_i + b^m) - 1 \geq 0$ where $m = 1, 2, \dots, M$ and w_i is the vertical vector of hyper plane, we can use the Lagrange method to search for the maximum value of the classification interval. That is to search the minimum of $L(w_i, \xi^i) = \frac{1}{2} \|w_i\|^2 + C \sum_{i=1}^N \xi^i$, where C is a penalty factor (the greater C , the greater punishment for wrong classification) and ξ^i is a relaxation factor reflecting the distance between x_i and hyper plane. Then we can get the equation $\sum_{i=1}^n y_i^m a_i^m = 0$. Under the conditions $\sum_{i=1}^n y_i^m a_i^m = 0$ and $a_i^m \geq 0$, the best value of a^m can be found by maximizing $H(a^m) = \sum_{i=1}^n a_i^m - \frac{1}{2} \sum_{i,j=1}^n a_i^m a_j^m y_i^m y_j^m K(x_i, x_j)$. Then substituting a feature vector belonging to the m -th class, say x_z , into (5) yields $f^m(x_z) = \text{sign}(\sum_i a_i^m y_i^m K(x_i, x_z) + b^m) = 1$, from which the value of b^m can be determined.

In the testing process, the decision function in (5) is used to determine whether a data object x belongs to the m -th class, according to the value of $f^m(x)$. Fig. 11 shows the flow chart of the testing process. The data object x is first tested by the SVM classifier 1 (SVM1). If $f^1(x) = 1$, the data object x is recognized as class 1. If not, it will be sent to classifier SVM2 to determine whether it belongs to class 2. Following the same procedure, the algorithm can find out which class the data object x belongs to. Clearly, in most cases, the number of classifiers needed to classify a data object is less than M . Hence this algorithm is more efficient in computation than the normal one-against-all method.

If $f^m(x) = -1$ for all $m = 1, 2, \dots, M$, it means that x belongs to a class that is different from the M classes given by the N training samples. In contrast, the normal one-against-all method will yield a wrong classification result for this data object.

4. Experimental results and analysis

4.1. Experimental results

The experiments were conducted on a straight road that is 2-km long and 12-m wide. The surface of the road is reinforced concrete. The sensor was properly installed to ensure that its sensitive axis was parallel with the directions of moving vehicles. The proposed vehicle detection and classification approach was tested using 93 vehicles, including 27 heavy tracked vehicles, 28 light tracked vehicles and 38 light wheeled vehicles. The vehicles passed the sensor within a distance of three meters to ensure that the

vehicles were in the detection range of the sensor. The sensor signal was sampled at the frequency of 2000 Hz. The structure of the detection system and the sensor installation diagram are shown in Figs. 2 and 3, respectively. During the test period, the weather was fine, with 30–35 °C temperature and 1–2 wind force level.

In the experiments, the features of concavity area, convexity area, the angle of concave part, and the angle of convex part of the waveform are extracted for vehicle detection and classification. The successful vehicle detection rate is 100%. Then the ISVM classifier which is based on the improved one-against-all algorithm is used to classify and recognize the vehicles. The results of vehicle detection and recognition using the ISVM classifier are shown in Table 1. Given the same extracted features, we also use other classifiers to classify and recognize vehicles, including the support vector machine (SVM), the back-propagation neural networks (BPNN) and the k-nearest neighbor (KNN) classifiers. The recognition rates on these classifiers are shown in Table 2.

4.2. Experimental analysis

Due to complex environment and some uncertain interference, errorless recognition cannot be guaranteed at this stage. We can see from Table 1 that the recognition rates of the proposed ISVM classifier are higher than 92% for the heavy tracked vehicles and light tracked vehicles, while its recognition rate on the light wheeled vehicles is 89.5%. The relatively lower recognition rate on the light wheeled vehicle result from the smaller size and volume

Table 1
Results of vehicle detection and recognition using the ISVM classifier.

	Heavy tracked vehicle	Light tracked vehicle	Light wheeled vehicle
Number of vehicles	27	28	38
Number of correct recognition	25	26	34
Number of incorrect recognition	2	2	4
Recognition rate (%)	92.6	92.86	89.5

Table 2
Recognition rates of SVM, BPNN and KNN classifiers.

Recognition rate	Heavy tracked vehicle (%)	Light tracked vehicle (%)	Light wheeled vehicle (%)
SVM classifier	88.9	85.7	84.2
BPNN classifier	81.5	78.6	76.3
KNN classifier	77.8	71.4	71.1

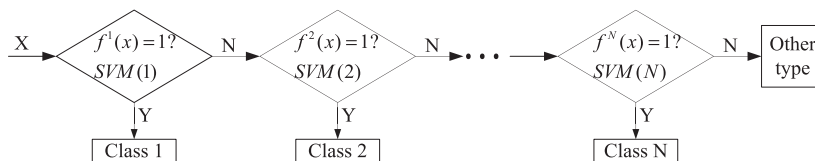


Fig. 11. Flow chart of the improved one-against-all algorithm.

of light wheeled vehicles, compared to the other two types of vehicles. Thus, the amplitudes of magnetic signals generated by the light wheeled vehicles are smaller. Consequently, when a light wheeled vehicle is not close to the sensor, the magnetic features about the concavity and convexity areas fall outside the normal range.

From Table 2, it can be seen that the recognition rates of the SVM classifier are higher than those of the BPNN and KNN classifiers, regardless of the types of vehicles. This is because the SVM classifier is robust to small sample set, which is the case in our experiments. Nevertheless, the ISVM classifier outperforms all the other classifiers as it obtains the highest recognition rates for all types of vehicles. This verifies the effectiveness of the ISVM classifier in vehicle recognition and classification.

5. Conclusion

This paper shows that the signal measured by a MEMS magnetic sensor contains useful features of the vehicle passing by the sensor. By exploiting the features of the magnetic signal, a novel approach is proposed to detect, classify and recognize the moving vehicles. This approach is simple as only one magnetic sensor is used and efficient in computation thanks to the usage of the ISVM classifier which is based on an improved one-against-all algorithm. Our experiments show that the recognition rate of the proposed approach is about 90%. Compared with the commercial vehicle detection devices, our experimental system can not only detect the passage of vehicles but also classify and recognize the types of vehicles with high recognition rate.

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